InterYard: Investigating E-Plants as Mediators for Attuning Human to Biorhythms in Future Cities

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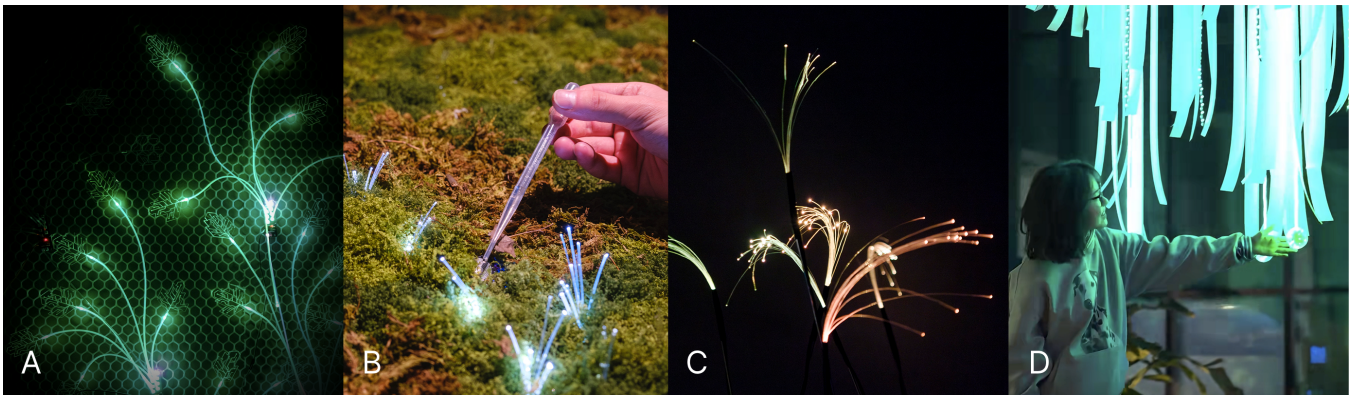


Figure 1: Four prototypes implemented in *InterYard*. A: *E-Ivy*. B: *E-Moss*. C: *E-Reed*. D: *E-Willow*.

Abstract

Rapid urbanization can detach residents from natural biorhythms, diminishing sensitivity to non-human life and ecological processes. Current Human-Nature Interaction (HNI) research often frames humans as authorities, struggling to align these rhythms. In response, we introduce *InterYard*, a post-anthropocentric artificial ecosystem featuring four bio-inspired *E-Plant* species: *E-Ivy*, *E-Moss*, *E-Reed*, and *E-Willow*. Equipped with environmental sensors and light expressions, each *E-Plant* emulates biological behaviors while reframing human interaction from one of dominance to participatory engagement. A preliminary study with 12 participants indicates that *InterYard* can meaningfully re-attune people to biorhythms, cultivate multispecies empathy, and deepen ecological awareness. This work contributes a novel HNI design paradigm and presents a speculative prototype for fostering ecological consciousness in future cities.

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CCS Concepts

• **Human-centered computing** → **Systems and tools for interaction design**; **HCI theory, concepts and models**.

Keywords

post-anthropocentric HCI, artificial ecology, human-nature interaction, more-than-human, ecological rhythms, tangible interaction, environmental sensing

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1 Introduction

Over-urbanization has reshaped not only physical landscapes but also profoundly altered how humans perceive nature [22]. The efficiency, immediacy and linearly-based logic of city planning is incongruous with gradual biological rhythms [8]. As a result, naturally-growing plants are often reduced from living organisms

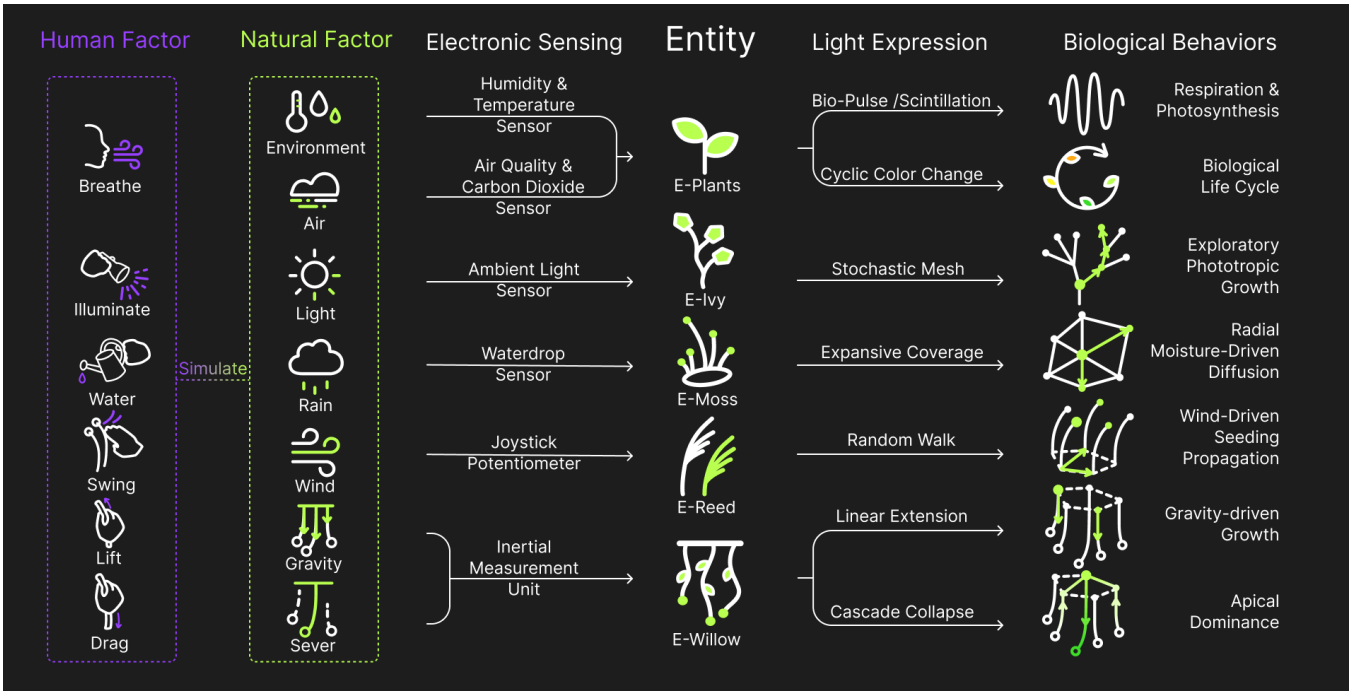


Figure 2: Design Space of *InterYard*: Translate human and natural factors through *E-Plants*' electronic sensing to designated light expressions according to diverse biological behaviors.

to static green decorations. This process diminishes residents' perception of natural cycles [25], reinforcing this disconnect between modern societal rhythms and biological rhythms [2, 4].

To address this, recent Human-Nature Interaction (HNI) studies have explored various technical paths [19]. Environmental data visualization employs sensors to render ecological signals as visual feedback [6, 30], while 3D scanning and XR technologies reconstruct natural environments for immersive experiences [15, 16]. Wearable and sensory-augmentation systems further explore embodied feedback to mediate human–nature relations [18, 24, 27, 32]. Other studies capture plant bioelectrical signals [11] and translate them into sound or visual media [9, 12, 16] to explore cross-species communication. While these studies have expanded the HNI field through making natural processes more perceptible or interactive [31], mainstream methods have defaulted to viewing plants as a trigger for human experience [5, 17, 28] instead of a system with rhythm of itself [23]. The construction of a long-term co-evolving system where humans are not the dominant controllers [3] remains an area to be explored [8].

To address the gap in this area, we suggest a systemic transition from "one-directional perception" to "symbiosis", coinciding with the post-humanist HCI trend of shifting design intent from "serving humans" to "reciprocable interaction between human and non-human entities" [10, 26]. We base this transition on two central principles: (1) The attunement of timescales, where the system can transform then sync timescales across species to bridge how humans perceive time by days versus plants' long term gradual growth [1, 2, 21], and (2) symbiotic participation, where human actions constitute as a variable input into the system alongside environmental variables [14, 20] as opposed to being the observer or dominator [7].

To enact this transition, we introduce *InterYard*: an artificial ecological system featuring *E-Plants* which centers on collecting environmental and human input through electronic sensing, translating abstract behaviors and gradual biorhythms into perceptible light expressions. As illustrated in Figure 1, we implement four concrete prototypes, *E-Ivy*, *E-Moss*, *E-Reed* and *E-Willow*. A preliminary qualitative evaluation was conducted through field deployment and semi-structured interviews to explore how interaction with *E-Plants* may support the perception of ecological rhythms and non-human timescales. Aligning the CHI 2026 theme of "Creating Tomorrow Together" with plants, this work contributes to:

- The design and implementation of *InterYard*, an artificial ecological embedded system translating abstract biological behaviors into perceptible light expressions.
- The novel proposal of a post-anthropocentric HNI design space that positions humans as participatory variables for perceiving biorhythms, supported by a preliminary evaluation (N=12).

2 InterYard System

As illustrated in Figure 2, the design space of *InterYard* centers on the translation of "Human/Natural Factor - Electronics - Light Expression". *E-Plants* are artificial plant-like entities that simulate plants' biological mechanisms with designated sensors through translating environmental and human input into various light

expressions. Notably, these embedded systems can operate autonomously, where human factors act as alternative variables, capable of influencing but not dominantly controlling *E-Plants*. The system conveys the growth and propagation of *E-Plants* through various light expressions [13, 29], such as rhythmic fluctuations and specific topological spread. These processes are visually enhanced and accelerated for month-long plant development to have immediate visual representations, fostering users' intuitive understanding of biorhythms. To demonstrate diverse biological behaviors within our design space, we present four *E-Plants* prototypes inspired by ivy, moss, reed and willow.

2.1 Common Growth Logic for All E-Plants

Every *E-plant* is capable of simulating plants' real-time respiration and photosynthesis, as well as the biological life cycle of growth.

Embedded sensors drive *E-plant* growth dynamics, including a DHT11[1] for temperature and humidity, a JW01-CO2[2] for CO_2 concentration, and a GP2Y1014AU0F[3] for PM2.5 levels. Warmer and more humid conditions accelerate development while degraded air quality will slow growth, triggering chromatic transitions which are rendered through WS2812 LEDs.

E-Plant growth states are amplified through dynamic light effects and color transitions. A complete life cycle, from seedling to maturity to withering, is represented by color shifts, progressing from bright green to dark green, then to yellow, and finally fading. To further enhance the dynamic effect of growth, light pulses mimic tidal movements, with effects described by the function $y = k_1 x + k_2 \sin(x)$. Human interaction can directly influence environmental parameters by breathing deeply to DHT11 – additional temperature and humidity increases k_1 for more vigorous growth, while CO_2 concentration simultaneously increases k_2 to simulate stronger respiration.

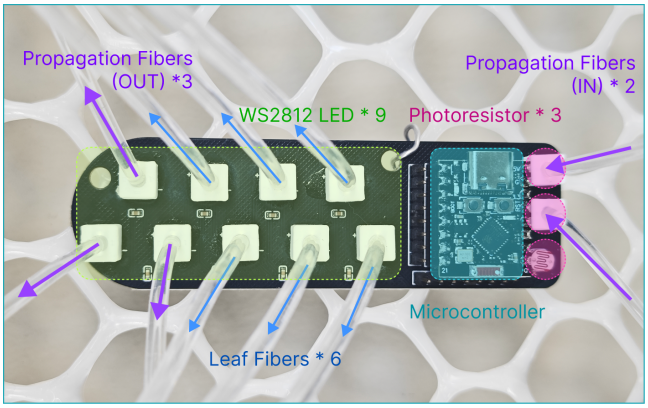


Figure 3: *E-Ivy*'s control board.

2.2 E-Ivy

E-Ivy is designed to visualize the phototropic biological behavior of plants.

For hardware, the primary visual elements of *E-Ivy* are optical fibers serving as vines, which illuminate laser-engraved acrylic "leaves" (see Figure 1.A) via connected WS2812 LEDs. Other leaf fibers transmit light from LED to the GL5528 photoresistor[4] of another *E-Ivy* node, forming an optical propagation link, as illustrated in Figure 3. This structure enables *E-Ivy* to process logic and relay signals. When a photoresistor is activated by light from prior *E-Ivy*, the connected node will sequentially light up six propagation fibers, then select one of three connecting fibers at random to activate the next level of *E-Ivy*.

E-Ivy is mounted on the wall, forming a Directed Acyclic Graph (DAG) tree structure, in which *E-Ivy* nodes serve as vertices and optical fibers act as edges. This network is a Finite State Machine (FSM) for *E-Ivy*'s potential growth. Each growth iteration first activates the initial *E-Ivy*, then randomly triggers a propagation event at each node. Users can guide or inhibit the propagation of *E-Ivy* in a specific direction by controlling photoresistors, such as manually by blocking light during the day or using a flashlight to guide growth at night.

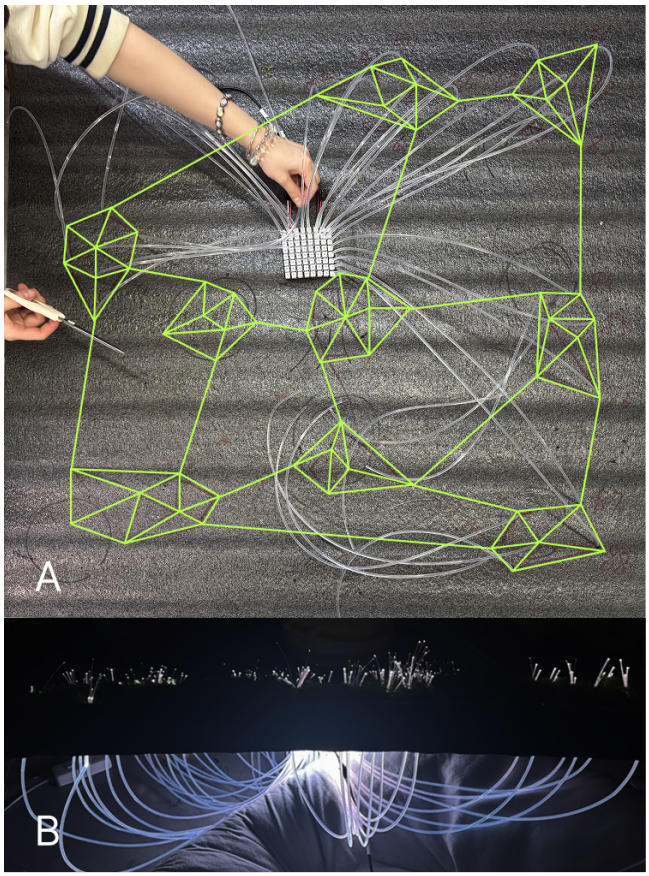


Figure 4: Implementation details of *E-Moss*. A: UCG structure for control strategy. B: Optical fibers connected to LED matrix.

[1] https://arduinogetstarted.com/tutorials/arduino-dht11
[2] https://sharingwin.com/product/jw01-co2-yys-co2-gas-sensor/
[3] https://www.adafruit.com/product/4649
[4] https://www.digikey.com/en/htmldatasheets/production/2014143/0/0/1/gl5528

2.3 E-Moss

E-Moss draws on the hygrotropic behavior of moss to attune people to moisture-driven plant propagation.

For hardware, the primary visual elements of *E-Moss* are short clustered thin optical fibers serving as moss strands (see Figure 1.B), which are inserted on a styrofoam surface. *E-Moss* points are not evenly distributed on the surface (see Figure 4.A), but mimic the clumped and discrete layout of real moss. The 64 scattered *E-Moss* points are assigned to an 8×8 matrix of WS2812 LEDs (see Figure 4.B). Resistive rainfall sensors[5] are evenly distributed on the surface to detect waterdrop events.

For interaction, each *E-Moss* point can be regarded as a vertex and connected to other surrounding *E-Moss* points through hypothetical edges, forming an Undirected Cyclic Graph (UCG) structure to simulate the potential growth space. When the rainfall sensor is activated, *E-Moss* will trigger light diffusion along adjacent edges, with the diffusion radius (number of edge steps in the UCG) mapped to the magnitude of resistance change; after water drops flow away from the sensor, the light effect will gradually fade. Users can guide the propagation of *E-Moss* in a specific direction by creating waterdrop events of different scales. Individuals can use a water dropper to activate local but intense propagation, or trigger widespread diffusion using a spray pot.

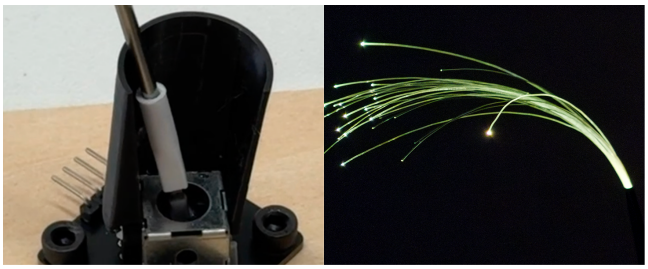


Figure 5: Base and tip structure of *E-Reed*.

2.4 E-Reed

E-Reed is designed to attune people to wind-driven propagation of plants.

E-Reeds are fixed on a laser-cut wooden frame and arranged to form a 2D distribution, where each *E-Reed* serves as a discrete point and the potential wind propagation paths are defined as vectors. Their primary visual elements are optical fiber plumes on the top of a thin metal rod stem (see Figure 5.B), which connect through a black optical fiber to a WS2812 LED at the root light source. The base of the flexible metal rod is attached to a joystick potentiometer[6] via a 3D printed part (see Figure 5.A), which maps the swaying of reeds to the X/Y resistance values which encode the vector's direction. The first derivative of the X/Y values maps to vector's modulus, guiding the strength of light propagation along the wind direction. Light expressions propagate along this vector, and pulse effects are used to simulate the swaying of reeds in the wind.

[5]https://arduinogetstarted.com/tutorials/arduino-rain-sensor
[6]https://www.adafruit.com/product/3102

Users can gently or vigorously swing the flexible metal rods to simulate different wind intensities and directions, thereby guiding the propagation of *E-Reeds*.



Figure 6: A: Control unit in the bulb at the tip of each LED strip. B: Light Effect enhanced by blurring materials.

2.5 E-Willow

E-Willow is designed to demonstrate the geotropism and apical dominance characteristics of plants.

For hardware, the primary visual element of *E-Willow* is two bonded WS2812 LED strips[7] hung from the ceiling, serving as branches to mimic the growth form of natural willows. An MPU6050 IMU[8] is embedded in the bulb (see Figure 6.A) at the tip of each LED strip (willow branch) to detect real-time acceleration and calculate gesture via Kalman filtering[9]. We further introduced "E-hormones" to refine the control logic to model light effects based on willow's apical dominance and lateral bud inhibition.

For interaction, the posture of the *E-Willow* branch tip is modeled by a 3D vector, whose Z-component can accelerate or inhibit growth. Shaking the branch tip is interpreted as a signal to accelerate growth, but vigorous shaking triggers "branch breakage", where light cascades and fades downward from the top, and lateral"buds" (LED segments previously inhibited by apical dominance) activate to form a new growth cycle. Users can guide the growth of *E-Willow* by orienting the branch tip, or digitally "break" the *E-Willow* branch to reset a new growth cycle.

3 PRELIMINARY EVALUATION

We deployed the system in an indoor school courtyard and invited 12 participants (P1-P12, aged 19–24) for interactive observation and semi-structured interviews, guided by open-ended questions to capture participants' shifting perception of non-human timescales. The study preliminarily verified the system's effectiveness in reconstructing the perception of ecological rhythms.

Rhythmic Perception via Temporal Folding. Participants (e.g., P4, P9) noted that observing the periodic the light patterns allowed them to visualize environmental data as biological movement. For instance, placing a hand near the sensors caused subtle fluctuations in the *E-Plants*' "light respiration", making invisible metabolic processes tangible.

From Control to Attunement. P1 and P8 initially sought buttons for immediate feedback, reflecting anthropocentric control.

[7]https://ledsuntech.com/ws2812-led-strip/
[8]https://invensense.tdk.com/products/motion-tracking/6-axis/mpu-6050/
[9]https://sparxeng.com/blog/software/imu-signal-processing-with-kalman-filter

The system's pace forced an attunement to the plant's rhythm; as P3 remarked: "I cannot command it to grow, it feels like a living being rather than a lighting fixture."

Cross-species Empathy. All 12 participants expressed affective resonance, ranging from surprise to care, demonstrating behaviors beyond object manipulation. P7, recalling a vine wall on their commute, remarked: "It was the first time I truly 'gazed' at a plant... I suddenly realized that these plants are lives almost equal to us".

4 CONCLUSION AND FUTURE WORK

Humans as a Participatory Variable. *InterYard* challenges the human-initiated interaction paradigm often seen in HNI. In this system, human behaviors (e.g., touch) are not prioritized commands but serve as a variable input alongside environmental factors. This shifts the user's role from a controller to a symbiotic participant. By positioning humans as just one part of the loop, the system fosters a deeper understanding of the limited human influence on natural systems, encouraging respect for non-human temporalities in urban spaces.

Speculative Prototype for Virtual Ecology. *InterYard* does not aim to provide a realistic substitute for nature; instead, it acts as a speculative prototype that distills biological logic and gradual biological rhythms. In the context of future cities, E-Plants serve as a cognitive tool that selectively amplifies natural cycles. This reconstruction bridges the gap between urban efficiency and biological time, proving the value of artificial ecological systems in promoting post-humanist reflection.

Future work will expand on the current *E-Plants* prototype: (1) *Spatio-temporal expansion*: Conducting long-term deployment in diverse public spaces to investigate whether extended exposure drives lasting shifts in users' perception of natural cycles. (2) *System agency enhancement*: Integrating Generative AI to foster self-evolving mechanisms, further testing the boundaries of empathy and coexistence between human and non-human systems.

In conclusion, this study explores a pathway to reconstruct the human-nature connection in highly artificial environments through *InterYard*. We propose a symbiotic interaction model where humans are defined as participatory variables within the ecosystem. By translating abstract biological behaviors, the system activates a re-perception of ecological rhythms. This offers a post-humanist design space for future cities: technology serves not only to reshape nature but to assist humans in learning how to coexist with non-human life.

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References

[1] Michelle Bastian. 2012. Fatally Confused: Telling the Time in the Midst of Ecological Crises. *Environmental Philosophy* 9, 1 (2012), 23–48. https://doi.org/10.5840/envirophil2012913

[2] Fiona Bell, Joshua Coffie, and Mirela Alistar. 2024. Bio-Digital Calendar: Attuning to Nonhuman Temporalities for Multispecies Understanding. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Cork, Ireland) *(TEI '24)*. Association for Computing Machinery, New York, NY, USA, Article 38, 15 pages. https://doi.org/10.1145/3623509.3633386

[3] Fadi Botros, Charles Perin, Bon Adriel Aseniero, and Sheelagh Carpendale. 2016. Go and Grow: Mapping Personal Data to a Living Plant. In *Proceedings of the International Working Conference on Advanced Visual Interfaces* (Bari, Italy) *(AVI '16)*. Association for Computing Machinery, New York, NY, USA, 112–119. https://doi.org/10.1145/2909132.2909267

[4] Patricia Ciobanu and Oskar Juhlin. 2022. Me, the Hill and My Browser – Investigating the Role of Time in Posthuman Interaction. In *Nordic Human-Computer Interaction Conference* (Aarhus, Denmark) *(NordiCHI '22)*. Association for Computing Machinery, New York, NY, USA, Article 71, 12 pages. https://doi.org/10.1145/3546155.3546651

[5] Donald Degraen, Hannah Hock, Marc Schubhan, Maximilian Altmeyer, Felix Kosmalla, and Antonio Krüger. 2022. FamilyFlower: an Artifical Flower to Foster Distant Family Connections. In *Proceedings of the 20th International Conference on Mobile and Ubiquitous Multimedia* (Leuven, Belgium) *(MUM '21)*. Association for Computing Machinery, New York, NY, USA, 204–207. https://doi.org/10.1145/3490632.3497833

[6] Tshering Dema, Margot Brereton, Jessica L. Cappadonna, Paul Roe, Anthony Truskinger, and Jinglan Zhang. 2017. Collaborative Exploration and Sensemaking of Big Environmental Sound Data. *Comput. Supported Coop. Work* 26, 4–6 (Dec. 2017), 693–731. https://doi.org/10.1007/s10606-017-9286-9

[7] Jan Fell, Travis Greene, Jyun-Cheng Wang, and Pei-Yi Kuo. 2020. Beyond Human-Centered Design: Proposing a Biocentric View on Design Research Involving Vegetal Subjects. In *Companion Publication of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven, Netherlands) *(DIS' 20 Companion)*. Association for Computing Machinery, New York, NY, USA, 209–214. https://doi.org/10.1145/3393914.3395883

[8] Marcus Foth and Glenda Amayo Caldwell. 2018. More-than-Human Media Architecture. In *Proceedings of the 4th Media Architecture Biennale Conference* (Beijing, China) *(MAB '18)*. Association for Computing Machinery, New York, NY, USA, 66–75. https://doi.org/10.1145/3284389.3284495

[9] Raune Frankjær and Maggie Kitel. 2016. Flora luma: a research-oriented artistic exploration of human-vegetal participatory space. In *Proceedings of the 14th Participatory Design Conference: Short Papers, Interactive Exhibitions, Workshops - Volume 2* (Aarhus, Denmark) *(PDC '16)*. Association for Computing Machinery, New York, NY, USA, 61–62. https://doi.org/10.1145/2948076.2948115

[10] Elisa Giaccardi, Johan Redström, and Iohanna Nicenboim. 2024. The making(s) of more-than-human design: introduction to the special issue on more-than-human design and HCI. *Human–Computer Interaction* 40, 1–4 (May 2024), 1–16. https://doi.org/10.1080/07370024.2024.2353357

[11] Peter A. Gloor. 2025. Plant Bioelectric Early Warning Systems: A Five-Year Investigation into Human-Plant Electromagnetic Communication. arXiv:2506.04132 [q-bio.OT] https://arxiv.org/abs/2506.04132

[12] Lone Koefoed Hansen and Margrete Lodahl Rolighed. 2024. Encountering Human-Plant Relations: a Discussion of How Interaction Design Can encourage Human Sensibility to More-than-human Plants. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) *(DIS '24)*. Association for Computing Machinery, New York, NY, USA, 1400–1411. https://doi.org/10.1145/3643834.3661586

[13] David Holstius, John Kembel, Amy Hurst, Peng-Hui Wan, and Jodi Forlizzi. 2004. Infotropism: living and robotic plants as interactive displays. In *Proceedings of the 5th Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques* (Cambridge, MA, USA) *(DIS '04)*. Association for Computing Machinery, New York, NY, USA, 215–221. https://doi.org/10.1145/1013115.1013145

[14] Yuan-Yao Hsu, Wenn-Chieh Tsai, Wan-Chen Lee, and Rung-Huei Liang. 2018. Botanical Printer: An Exploration on Interaction Design with Plantness. In *Proceedings of the 2018 Designing Interactive Systems Conference* (Hong Kong, China) *(DIS '18)*. Association for Computing Machinery, New York, NY, USA, 1055–1068. https://doi.org/10.1145/3196709.3196809

[15] Youyang Hu, Cyprien Fol, Chiaochi Chou, Verena Griess, and Yasuaki Kakehi. 2024. Eye of Flora: Encountering Nature through the Mixed Reality Lens of Plant-Environment Interactions. In *SIGGRAPH Asia 2024 Art Papers (SA '24)*. Association for Computing Machinery, New York, NY, USA, Article 16, 6 pages. https://doi.org/10.1145/3680530.3695442

[16] Youyang Hu, Cyprien Raymi Fol, Chiaochi Chou, Verena C Griess, and Yasuaki Kakehi. 2024. Immersive Flora: Re-Engaging with the Forest through the Visualisation of Plant-Environment Interactions in Virtual Reality. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) *(CHI EA '24)*. Association for Computing Machinery, New York, NY, USA, Article 407, 6 pages. https://doi.org/10.1145/3613905.3648675

[17] Sungjae Hwang, Kibeom Lee, and Woonseung Yeo. 2009. Introducing a current-based interactive plant. In *ACM SIGGRAPH ASIA 2009 Posters* (Yokohama, Japan) *(SIGGRAPH ASIA '09)*. Association for Computing Machinery, New York, NY, USA, Article 7, 1 pages. https://doi.org/10.1145/1666778.1666785

[18] Sylvia Janicki, Alexandra Teixeira Riggs, Noura Howell, Anne Sullivan, and Nassim Parvin. 2024. Sensing Bodies: Engaging Postcolonial Histories through More-than-Human Interactions. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Cork, Ireland) *(TEI '24)*. Association for Computing Machinery, New York, NY, USA, Article 41, 15 pages. https://doi.org/10.1145/3623509.3633389

[19] Madlen Kneile, Judith Dörrenbächer, Marc Hassenzahl, and Matthias Laschke. 2025. Mediating Human–Nature Relations through Technology: A Scoping Review of Post-Anthropocentric Artifacts and Their Design Strategies. *ACM Trans. Comput.-Hum. Interact.* 32, 5, Article 51 (Oct. 2025), 51 pages. https://doi.org/10.1145/3743674

[20] Satoshi Kuribayashi, Yusuke Sakamoto, Maya Morihara, and Hiroya Tanaka. 2007. Plantio: an interactive pot to augment plants' expressions. In *Proceedings of the International Conference on Advances in Computer Entertainment Technology* (Salzburg, Austria) *(ACE '07)*. Association for Computing Machinery, New York, NY, USA, 282–283. https://doi.org/10.1145/1255047.1255126

[21] Siân Lindley, Robert Corish, Elsa Kosmack Vaara, Pedro Ferreira, and Vygandas Simbelis. 2013. Changing perspectives of time in HCI. In *CHI '13 Extended Abstracts on Human Factors in Computing Systems* (Paris, France) *(CHI EA '13)*. Association for Computing Machinery, New York, NY, USA, 3211–3214. https://doi.org/10.1145/2468356.2479649

[22] Karina J Linnell and Serge Caparos. 2020. Urbanisation, the arousal system, and covert and overt attentional selection. *Current Opinion in Psychology* 32 (2020), 100–104. https://doi.org/10.1016/j.copsyc.2019.07.030 Socio-Ecological Psychology.

[23] Susan Loh, Marcus Foth, and Yasu Santo. 2024. The more-than-human turn in human-plant interaction design: From utilitarian object to living co-inhabitant. *Int. J. Hum.-Comput. Stud.* 181, C (Jan. 2024), 15 pages. https://doi.org/10.1016/j.ijhcs.2023.103128

[24] Takuya Mikami and Jun Fujiki. 2022. Development of a method to improve the empathy of observers by virtual experience of becoming a plant. *The Journal of the Society for Art and Science* 21, 2 (Jun 2022), 37–45. https://doi.org/10.3756/artsci.21.37

[25] Aurélien Miralles, Michel Raymond, and Guillaume Lecointre. 2019. Empathy and compassion toward other species decrease with evolutionary divergence time. *Scientific Reports* 9, 1 (2019), 1–10.

[26] Iohanna Nicenboim, Doenja Oogjes, Heidi Biggs, and Seowoo Nam. 2025. Decentering Through Design: Bridging Posthuman Theory with More-than-Human Design Practices. *Human-Computer Interaction* 40, 1-4 (2025).

[27] Stijn Ossevoort and Miguel Bruns. 2023. Embodying Wind through Flare: How Natural Phenomena Can Contribute to Enriching the Design of Interactive Systems. In *Proceedings of the Seventeenth International Conference on Tangible, Embedded, and Embodied Interaction* (Warsaw, Poland) *(TEI '23)*. Association for Computing Machinery, New York, NY, USA, Article 12, 11 pages. https://doi.org/10.1145/3569009.3572743

[28] Ivan Poupyrev, Philipp Schoessler, Jonas Loh, and Munehiko Sato. 2012. Botanicus Interacticus: interactive plants technology. In *ACM SIGGRAPH 2012 Emerging Technologies* (Los Angeles, California) *(SIGGRAPH '12)*. Association for Computing Machinery, New York, NY, USA, Article 4, 1 pages. https://doi.org/10.1145/2343456.2343460

[29] Benjamin John Southwell, Yin-Lee Ho, Frederic Anthony Robinson, and David Gunawan. 2025. Creating and Delivering Transferable Object-Based Light Content. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*. Association for Computing Machinery, New York, NY, USA, Article 158, 7 pages. https://doi.org/10.1145/3706599.3719694

[30] Martin Tironi. 2024. Hybrid Ecologies of artificial intelligence: prototyping terrestrial practices through a design installation. *Proceedings of DRS* (2024). https://api.semanticscholar.org/CorpusID:270557616

[31] Sarah Webber, Ryan M. Kelly, Greg Wadley, and Wally Smith. 2023. Engaging with Nature through Technology: A Scoping Review of HCI Research. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) *(CHI '23)*. Association for Computing Machinery, New York, NY, USA, Article 521, 18 pages. https://doi.org/10.1145/3544548.3581534

[32] Takashi Yamauchi, Jinsil Hwaryoung Seo, and Annie Sungkajun. 2018. Interactive Plants: Multisensory Visual-Tactile Interaction Enhances Emotional Experience. *Mathematics* 6, 11 (2018). https://doi.org/10.3390/math6110225